

Project Design Phase -II

Solution Requirements (Functional & Nonfunctional)

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Project Name	Automated Network Request Management System
Maximum Marks	4 marks

1. System Functional Requirements

Use the table below to detail the core platform constraints, data sanitization rules, and automated logic flows running inside your workspace instance.

Requirement ID	Module / Area	Requirement Type	Description	Acceptance Criteria	Priority	Status
REQ-NW-001	Service Portal Intake	Functional	The system shall present a standardized, single point of ingestion via a custom Service Portal Catalog Item for all internal network request actions.	Given an authenticated corporate employee accesses the portal, when they select the network form, then it must load matching corporate design schemas with clear fields.	Critical	Proposed
REQ-NW-002	Session Auto-Population	Functional	The intake form shall automatically query the foundational user profile table (sys_user) to map user session properties into non-editable metadata blocks.	Given the form loads, when the system context reads the active login ID, then it must auto-fill employee name, corporate email, and department codes instantly.	High	Proposed
REQ-NW-003	Client-Side Sanitization	Functional	The catalog form variables shall utilize client-side UI Policies ,Catalog Client Scripts to enforce strict alphanumeric text syntax constraints.	Given an asset string input (e.g., VLAN tags or IP subnets), when the user inputs invalid text or formats, then the system must block submission and render error flags.	Critical	Proposed

REQ-NW-004	Flow Designer Approvals	Functional	The backend system shall trigger automated multi-stage routing loops using Flow Designer hooks upon transaction ingestion.	Given a successfully validated submission, when the workflow engine checks manager hierarchies via relational tables, then it must instantly route interactive approval cards.	High	Proposed
REQ-NW-005	Workspace Queue Management	Functional	The platform shall split approved parent entries into distinct engineering task cards (u_network_task) routed to the network squad workspace.	Given manager authorization event signals, when the background script engine parses variables, then it must spawn child fulfillment cards containing 100% complete metrics.	High	Proposed
REQ-NW-006	Custom Ledger Recording	Functional	The system shall commit all final transactional parameter updates to a custom relational tracking directory table (u_network_database) upon task closure.	Given an analyst marks an engineering task card as closed, when business rules fire, then the system must execute write parameters to update persistent schemas.	Medium	Proposed
REQ-NW-007	Portal Progress Indicator	Functional	The user dashboard shall display an asynchronous visual stage tracking bar highlighting active transaction progress metrics.	Given a request ticket lifecycle state alteration, when backend workflow flags shift, then the portal header progress bar must update its view dynamically.	Medium	Proposed
REQ-NW-008	Audit Trail Compliance	Functional	The application shall enforce platform-native tracking mechanisms to catch all security modifications, approvals, and system events.	Given any write action or state change on target fulfillment cards, when the query executes, then the database must write unalterable history logs to sys_audit records.	High	Proposed

2. Non-Functional Requirements (NFR)

Use the table below to map the operational guardrails, performance benchmarks, and compliance standards required for system production readiness.

Requirement ID	Module / Area	Requirement Type	Description	Acceptance Criteria	Priority	Status
REQ-NFR-001	Performance & Speed	Non-Functional	Client-side UI Scripts, Catalog Client Scripts, and UI Policies must process configurations with minimal latency tails.	Total page interaction and form validation runtime must not introduce more than 1.5 seconds of total lag during data entry.	High	Proposed
REQ-NFR-002	Security & RBAC	Non-Functional	Access to custom data structures and fulfillment processing pipelines must follow the Principle of Least Privilege.	Only users within the active Network Assignment Group have write permissions on u_network_task; end-users are restricted to read-only access.	Critical	Proposed
REQ-NFR-003	Reliability & Uptime	Non-Functional	The automated request platform components must remain consistently operational alongside the main corporate ServiceNow instance environment.	The module components must meet an active availability threshold of 99.9% uptime during standard enterprise operating windows.	High	Proposed
REQ-NFR-004	Scalability Framework	Non-Functional	The custom relational data directories and catalog data schemas must sustain increasing traffic volumes seamlessly.	The custom u_network_database directory must scale smoothly to handle up to 10,000 concurrent records without slowing down indexing lookup speeds.	Medium	Proposed
REQ-NFR-005	Data Integrity & Lock	Non-Functional	Automated database record transactions must lock values properly during writes to prevent missing context data configurations.	System business rules must maintain absolute transactional state sync loops, ensuring zero variable drops when moving parent details into child	Critical	Proposed

				cards.		
REQ-NFR-006	Audit Compliance	Non-Functional	The system must provide complete tracking history accountability for all infrastructure alterations to pass internal compliance checks.	100% of state change modifications, structural approvals, and transaction logs must be permanently saved inside immutable sys_audit streams.	High	Proposed